Ofqual QAN	**600/0406/X**
Quartz ID	**1156 / 1157**
Qualification	**Level 4 Diploma in Computing / Level 4 Diploma in Computing (with Business Management)**

Unit ID	**2223**
Assessment	**Computer Systems 2.0**
Assessment Type	**Global Examination**
Date	**Wednesday 4th June 2025**
Time	**2.00pm – 5.00pm**

Examination Paper

Answer ALL questions.

Clearly cross out any surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions
Do NOT copy the questions to your answer booklet

Marks

Question 1

For each answer, only provide the corresponding numeral (e.g., i, ii, iii, or iv).

a) Which component of the computer system is primarily responsible for temporarily holding data **and** instructions for quick access during processing? **1**

i) Cache
ii) Memory Unit
iii) Registers
iv) Random Access Memory

b) In data communications, which of the following is **NOT** considered a goal? **1**

i) Memory capacity
ii) Cost-effectiveness
iii) Word length
iv) Pipelining

c) What does SMTP stand for in email communication? **1**

i) Secure Mail Transfer Protocol
ii) Secure Messaging Transfer Protocol
iii) Simple Mail Transfer Protocol
iv) Simple Messaging Transfer Protocol

d) Which of the following best describes a fundamental characteristic of quantum computing? **1**

i) Quantum computers use classical bits (1s and 0s) to process information.
ii) Quantum computers rely on quantum bits (qubits) that can exist in multiple states simultaneously.
iii) Quantum computers are slower than traditional computers.
iv) Quantum computers are limited to solving only mathematical problems.

e) What is the primary benefit of CPU pipelining in computer architecture? **1**

i) It increases the clock speed of the CPU.
ii) It minimises the power consumption of the CPU.
iii) It allows multiple instructions to be processed simultaneously in different stages.
iv) It reduces the size of the CPU.

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Marks

f) Which of the following logic gates produces an output of ONE (1) only when both inputs are different? **1**

i) AND Gate

ii) OR Gate

iii) XOR Gate

iv) NOT Gate

g) What is the decimal equivalent of the binary number 1011? **1**

i) 8

ii) 9

iii) 10

iv) 11

h) Which type of CPU addressing mode uses a register's value as a pointer to the memory location of data? **1**

i) Indirect Addressing

ii) Immediate Addressing

iii) Direct Addressing

iv) Indexed Addressing

i) Which device is used to connect multiple computers within a local area network (LAN) and manage data traffic between them? **1**

i) Hub

ii) Switch

iii) Modem

iv) Router

j) Which system software is responsible for translating high-level programming language into machine code? **1**

i) SQL Database

ii) Operating System

iii) Interpreter

iv) Compiler

Total 10 Marks

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Marks

Question 2

a) Explain in detail the distinction between hardware and software in a computer system. Use examples to illustrate your answer. **5**

b) Discuss the function of the control unit in relation to the Central Processing Unit. **5**

Total 10 Marks

Question 3

Operating Systems are the 'backbone' of modern computing, facilitating the interaction between the hardware and software components of a computer system.

a) Discuss the following THREE (3) functions of an operating system with a description of their roles and how each interacts with hardware and software applications. **9**

- File System Management
- Device Management
- Security and Access Control

b) State ONE (1) additional *main* function of the OS. **1**

Total 10 Marks

Question 4

Computer programming is ONE (1) of the core elements of computing and understanding how instructions are processed and executed by a computer is fundamental.

a) Explain the concept of addressing **and** describe the differences between direct, immediate and indirect addressing. Provide at least ONE (1) example of how these modes differ. **5**

b) Explain the key differences between RISC (Reduced Instruction Set Computing) and CISC (Complex Instruction Set Computing) architectures. Highlight ONE (1) distinct advantage and ONE (1) distinct disadvantage for each architecture by focusing on the design principles or operational impact. **4**

c) Which architecture in b) would the processor of a new smartphone likely utilise? **1**

Total 10 Marks

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Marks

Question 5

a) Computer systems process binary numbers which can be expressed in different formats. **3**

Add binary numbers 10101 and 00100. Give the result in base 2. Show your working.

b) Convert the hexadecimal number F8D4 to its decimal value. Show your working. **3**

c) Represent the number 1874 in 2's complement. Show your working. **3**

d) Which number system is also known as the base-10 system? **1**

Total 10 Marks

Question 6

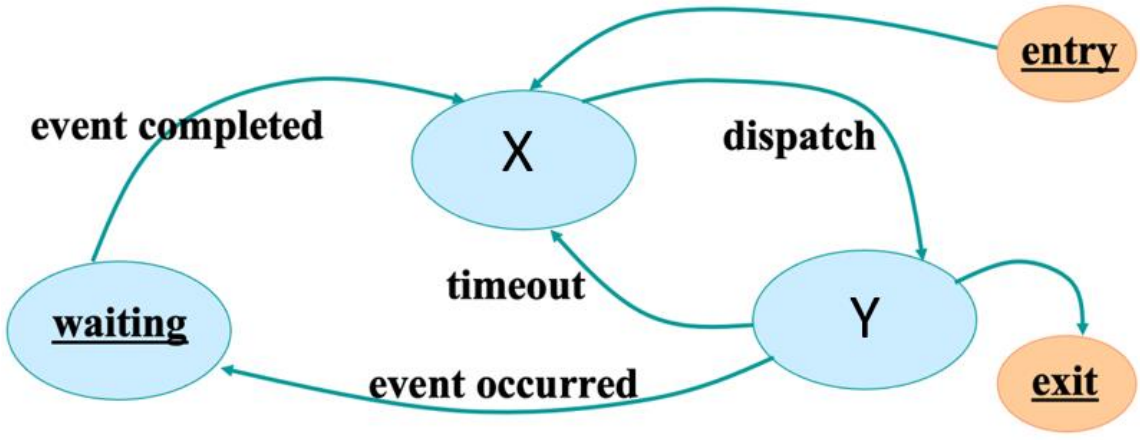
a) Explain the key differences between compilers and interpreters in terms of how they can help execute programs. For each, highlight how these differences affect their functionality. **4**

b) What is system software? Give TWO (2) key differences between system software and application software. **4**

In your answer, you should focus on each software's *purpose* and *dependency*.

c) Processes in a computer system can be in ONE(1) of several states. **2**

In the following diagram, state the missing states (denoted by X and Y).



Total 10 Marks

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Marks

Question 7

a) Name the THREE (3) main stages of a CPU's instruction cycle. **3**

b) Explain the purpose of an opcode. **2**

c) Explain the purpose of an operand in a CPU. Give ONE (1) example of an operation involving an operand. **3**

d) Before the CPU processes any data, it is checked via a parity checker. **2**

Describe how a parity checker works.

Total 10 Marks

Question 8

a) Is the following statement TRUE or FALSE? **1**

In a multi-core computer system, each core can execute independent tasks simultaneously.

b) A CPU, with its pipeline switched off, executes 2500 instructions in 5000 clock cycles. What is the average clock cycle per instruction executed? **1**

c) Outline FIVE (5) distinct characteristics of distributed computing. **5**

d) Micro-controllers are widely used in a number of different areas. Provide THREE (3) examples of application areas where micro-controllers are used. **3**

Total 10 Marks

Question 9

a) Explain what is meant by the term network topology. Provide TWO (2) examples of different network topologies and state their *fault tolerance* and *performance* characteristics. **7**

b) State how a router differs from a switch in terms of functionality. **2**

c) Name the widely adopted communication protocol responsible for allowing emails to be sent across the internet. **1**

Total 10 Marks

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Marks

Question 10

a) What is the octal equivalent of the hexadecimal value B35? Show your working. **3**

b) Complete the following table for the logic function: A.(B+C) **7**

A	***B***	***C***	***B + C***	**A . (B + C)**
0	*0*	*0*		
0	*0*	*1*		
0	*1*	*0*		
0	*1*	*1*		
1	*0*	*0*		
1	*0*	*1*		
1	*1*	*0*		
1	*1*	*1*	1	1

Total 10 Marks

End of paper

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